GSE 518
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Problem Set 6

You must turn in problems: 1, 2, 4, and 6(b). However, you are strongly encouraged to work on all the problems before the midterm. Please let me know if you have any questions.

1. Let the random variables X and Y have joint cdf $F(x,y) = (1-e^{-x})(1-e^{-y})$. The set of values for which the joint pdf is positive (i.e., the support of (X,Y)) is $\{(x,y) : x > 0, y > 0\}$.

   (a) Find the joint pdf, $f(x,y)$.

   (b) Are X and Y independent?

2. Let the random variables X and Y have joint pdf $f(x,y) = 0.25(2x+y)$ for $0 < x < 1$, $0 < y < 2$, and zero otherwise.

   (a) Find the probability $\Pr(X < 0.25, Y < 1)$.

   (b) Find the marginal distributions $f_X(x)$ and $f_Y(y)$.

   (c) Find $E(X)$.

   (d) Are X and Y independent?

   (e) Find the conditional density $f_{Y|X}(y|X=x)$.

   (f) Find the conditional density $f_{X|Y}(x|Y=y)$.

   (g) Find the probability $\Pr(X < 0.25|Y=1)$.

   (h) Find $E(X|Y=y)$.

   (i) Write down how you would calculate $E[E(X|Y=y)]$ (i.e., write down the integral you would have to calculate, but do not solve it).

   (j) In this example calculating the integral in (i) can be time consuming. Use one of the results seen in class to find $E[E(X|Y=y)]$.

3. (a) For random variables X and Y, and constant $a \in R$, show that $Cov(aX,Y) = aCov(X,Y)$. (*Hint:* You may want to start by using the definition of covariance, say, $Cov(U,V) = E\{[U-E(U)][V-E(V)]\}$.)

   (b) (*Just for general knowledge. In other words, you do not need to solve this one. It is very similar to the one in part (a) but involves more algebra.*) For random variables X, Y, W and Z; and constants $a, b, c, d \in R$, show that

   $$Cov(aW+bX, cY+dZ) = acCov(W,Y)+adCov(W,Z)+bcCov(X,Y)+bdCov(X,Z)$$

4. Let X and Y follow a bivariate normal distribution with parameters $\mu_x, \mu_y, \sigma_x^2, \sigma_y^2$ and ρ_{xy}. If $U = X+Y$ and $V = X-Y$, find an expression for ρ_{UV} (i.e., the correlation between U and V) as a function of $\mu_x, \mu_y, \sigma_x^2, \sigma_y^2$ and ρ_{xy}. (*Hint:* You may find the result in question 3(b) useful to calculate $Cov(U,V)$.)

5. Suppose the random variables $X_1, X_2, \ldots, X_n$ are mutually independent, and the distribution of X_i is a $Chi - squared(p_i)$. Let $Y = \sum_{i=1}^{n} X_i$. What is the distribution of Y? (*Hint:* You may take the mgf of a $Chi - squared(p_i)$ as given and apply one of the "implications from independence" we saw in class).

6. Let X and Y follow a bivariate normal distribution with parameters $\mu_x, \mu_y, \sigma_x^2, \sigma_y^2$ and ρ.

   (a) Show that X and Y are independent if and only if $\rho = 0$.

   (b) For any constants a and b, show that the distribution of $aX + bY$ is

   $$N\left(a\mu_x + b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2 + 2ab\rho\sigma_x\sigma_y\right)$$

   (*Note:* You can take as given the fact that linear combinations of normally distributed random variables are normal. In other words, in this problem you only have to verify the mean and variance of the random variable $aX + bY$).